

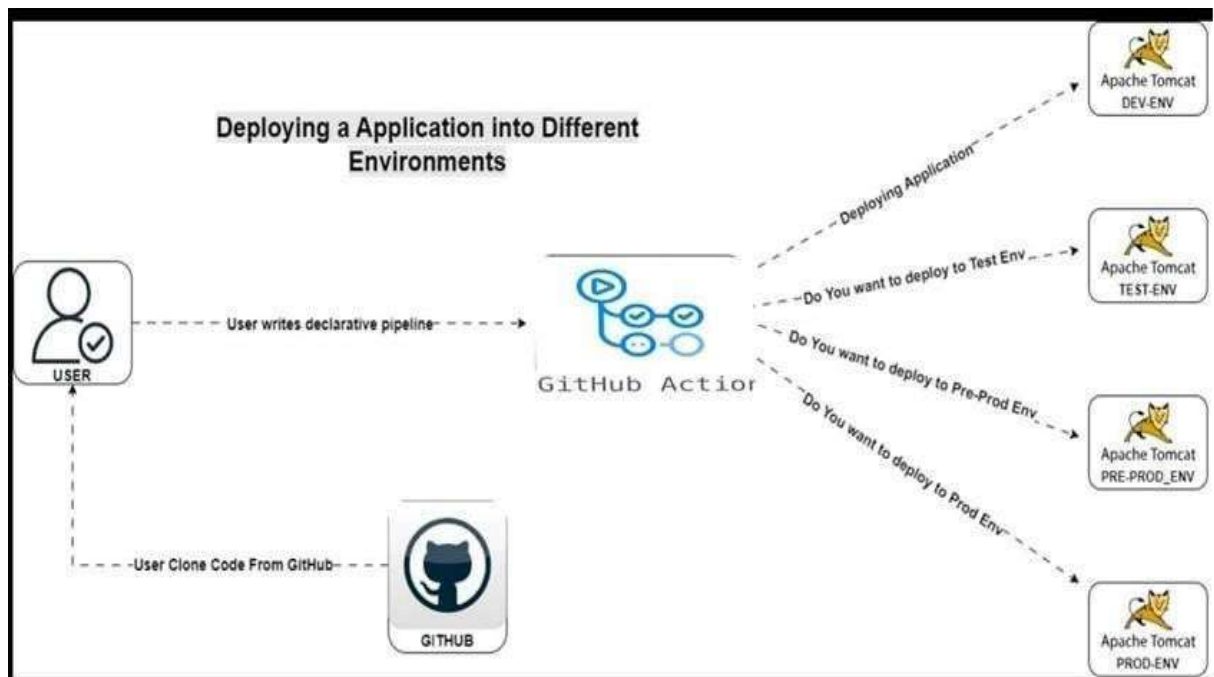
PROJECT – 4

Implemented a CI/CD pipeline with GitHub Actions to deploy applications on Apache Tomcat

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Batch:147

DEPLOY APPLICATION TO TOMCAT USING CI/CD PIPELINE WITH GITHUB ACTIONS



Tools Used:

- ☐ GitHub – To store source code and run GitHub Actions.
- ☐ Maven – To build the Java project and create the WAR file.
- ☐ Apache Tomcat – Application server to run the web app.
- ☐ AWS EC2 – Cloud VM used to host Tomcat.
- ☐ GitHub Actions – CI/CD tool used to automate build and deployment.

Introduction:to implemented a CI/CD pipeline with GitHub Actions to deploy applications on Apache Tomcat.

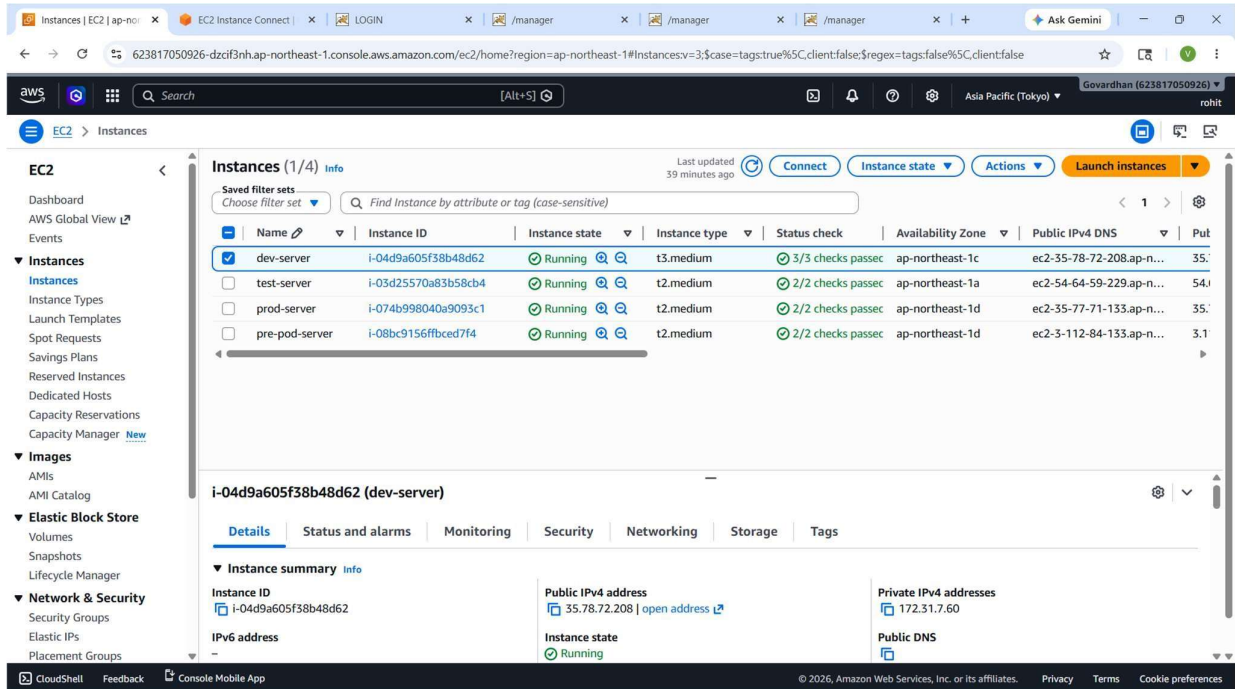
Automation has become a core component of modern software development, enabling faster releases, better code quality, and reliable deployment processes. Continuous Integration and Continuous Deployment (CI/CD) pipelines help development teams automate building, testing, and deploying applications, reducing manual involvement. This project demonstrates the creation of a CI/CD pipeline using GitHub Actions to automatically deploy a Java web application to an Apache Tomcat server. The main goal of this project is to develop an automated workflow that activates whenever new code is pushed to the GitHub repository. The pipeline carries out multiple stages, including retrieving the source code, configuring the Java environment, building the application with Maven, executing test cases, packaging the application as a WAR file, and deploying it to the Tomcat server.

By implementing GitHub Actions, the deployment process becomes consistent, repeatable, and significantly more efficient than manual deployment methods. Apache Tomcat is a widely adopted and lightweight application server designed for hosting Java-based web applications. In this project, the Tomcat Manager API is used to deploy the WAR file directly from the CI/CD pipeline. This removes the need for manual deployment through the Tomcat web console, helping to minimize errors and reduce deployment time. Security is maintained by using GitHub Secrets to store sensitive data such as server login credentials, host information, and deployment endpoints. This approach ensures that confidential details are protected while allowing secure communication between the CI/CD pipeline and the Tomcat server. In summary, this project illustrates how automation improves the software delivery lifecycle. By integrating GitHub Actions with Apache Tomcat, the deployment process becomes smooth, reliable, and efficient. This CI/CD solution accelerates development, improves stability, and follows professional, industry-standard deployment practices suitable for real-world applications.

PROCESS OF DEPLOY WAR FILE IN TOMCAT USING GITHUB ACTIONS

Step1: CREATE EC2 INSTANCES

- ☐ Create four EC2 instances to represent different Tomcat environments.
- ☐ Name each instance according to its environment, such as Dev, Test, PreProd, and Prod.



STEP – 2:

INSTALL JAVA AND MAVEN INSTALL and also TOMCAT ON EC2 Connect to DEV-Server in gitbash.

- ☐ Install the tomcat 9 in DEV-Server
- ☐ Apache Tomcat was downloaded, extracted, and configured to run on port 8080
- ☐ The tomcat-users.xml file was updated to create a user with manager-gui and manager-script roles.

```

root@ip-172-31-31-170: ~/tomcat/bin
root@ip-172-31-31-170:~# wget https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113.tar.gz
--2025-12-19 10:39:29-- https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113.tar.gz
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13049663 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.113.tar.gz'

apache-tomcat-9.0.113.tar.gz      100%[=====] 12.44M --.-KB/s   in 0.03s

2025-12-19 10:39:31 (358 MB/s) - 'apache-tomcat-9.0.113.tar.gz' saved [13049663/13049663]

root@ip-172-31-31-170:~# ls
apache-tomcat-9.0.113.tar.gz snap
root@ip-172-31-31-170:~# tar -xf apache-tomcat-9.0.113.tar.gz
root@ip-172-31-31-170:~# ls
apache-tomcat-9.0.113 apache-tomcat-9.0.113.tar.gz snap
root@ip-172-31-31-170:~# mv apache-tomcat-9.0.113 tomcat
root@ip-172-31-31-170:~# ls
apache-tomcat-9.0.113.tar.gz snap tomcat
root@ip-172-31-31-170:~# cd tomcat/
root@ip-172-31-31-170:~/tomcat# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
root@ip-172-31-31-170:~/tomcat# cd bin/

```

```

root@ip-172-31-31-170: ~/tomcat/bin
Using CATALINA_BASE: /root/tomcat
Using CATALINA_HOME: /root/tomcat
Using CATALINA_TMPDIR: /root/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /root/tomcat/bin/bootstrap.jar:/root/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@ip-172-31-31-170:~/tomcat/bin# cd
root@ip-172-31-31-170:~/tomcat# cd tomcat/
root@ip-172-31-31-170:~/tomcat# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
root@ip-172-31-31-170:~/tomcat# cd conf/
root@ip-172-31-31-170:~/tomcat/conf# ls
catalina catalina.properties jaspic-providers.xml logging.properties tomcat-users.xml web.xml
catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-users.xsd
root@ip-172-31-31-170:~/tomcat/conf# vi tomcat-users.xml
root@ip-172-31-31-170:~/tomcat/conf#
root@ip-172-31-31-170:~/tomcat/conf# cd
root@ip-172-31-31-170:~/tomcat# cd bin/
root@ip-172-31-31-170:~/tomcat/bin# ls
bootstrap.jar ciphers.bat configtest.bat digest.sh setclasspath.sh startup.sh tool-wrapper.sh
catalina-tasks.xml ciphers.sh configtest.sh digest.sh shutdown.bat tomcat-juli.jar version.bat
catalina.bat commons-daemon-native.tar.gz daemon.sh makebase.sh shutdown.sh tomcat-native.tar.gz version.sh
catalina.sh commons-daemon.jar digest.bat setclasspath.bat startup.bat tool-wrapper.bat
root@ip-172-31-31-170:~/tomcat/bin# ./shutdown.sh
Using CATALINA_BASE: /root/tomcat
Using CATALINA_HOME: /root/tomcat
Using CATALINA_TMPDIR: /root/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /root/tomcat/bin/bootstrap.jar:/root/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS: --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED
root@ip-172-31-31-170:~/tomcat/bin# ./startup.sh
Using CATALINA_BASE: /root/tomcat
Using CATALINA_HOME: /root/tomcat
Using CATALINA_TMPDIR: /root/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /root/tomcat/bin/bootstrap.jar:/root/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@ip-172-31-31-170:~/tomcat/bin#

```

STEP – 3: CREATE AMI

- ☐ Create an AMI Using DEV-Server
- ☐ using that AMI launch 3 -Instances for different environments. ☐ Name them as test, Pre-prod, Prod.
- ☐ Now refresh the ec2 instances you will get 3 instances adding.

The screenshot shows the AWS Management Console interface for Amazon Machine Images (AMIs). The left sidebar contains navigation links for various AWS services. The main content area displays the 'Amazon Machine Images (AMIs) (1/1) info' page. A table lists the available AMIs, with one entry: 'project-4-ami' (AMI ID: ami-04028e7694a4084f9). Below the table, the 'Details' tab for the selected AMI is expanded, showing information such as the image type (machine), platform details (Linux/UNIX), architecture (x86_64), and root device name (/dev/sda1).

Name	AMI name	AMI ID	Source	Owner	Visibility
☒	project-4-ami	ami-04028e7694a4084f9	623817050926/project-4-ami	623817050926	Private

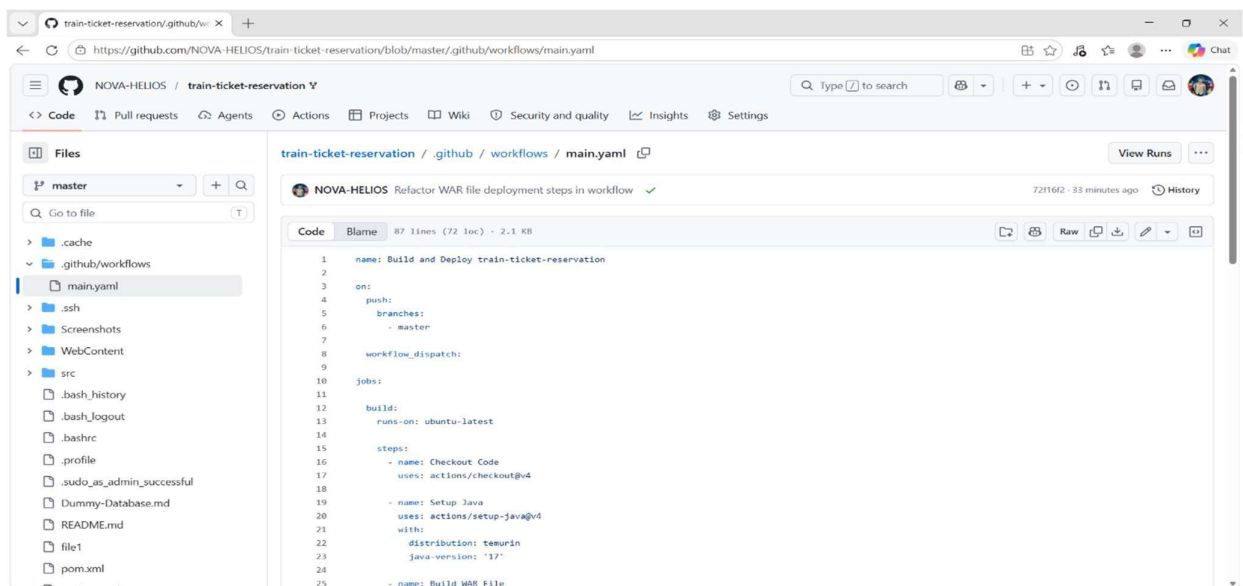
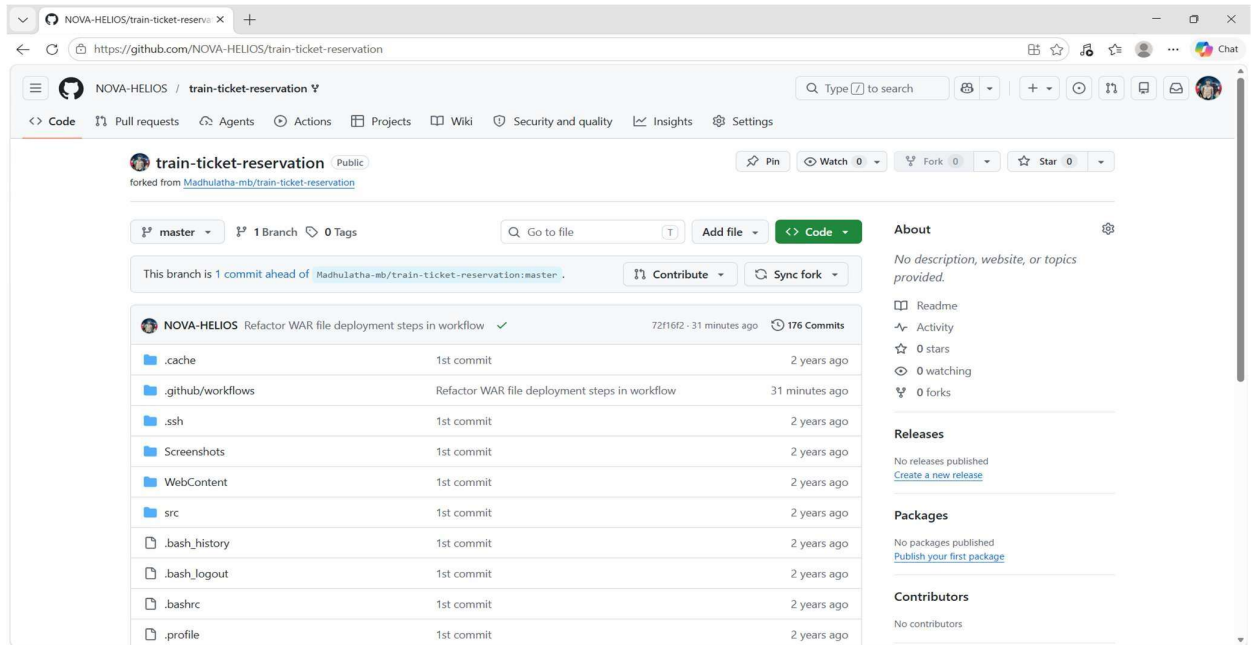
AMI ID: ami-04028e7694a4084f9

Image details		Platform details		Root device type	
AMI ID	ami-04028e7694a4084f9	Image type	machine	Platform details	Linux/UNIX
AMI name	project-4-ami	Owner account ID	623817050926	Architecture	x86_64
Root device name	/dev/sda1	Status	Available	Source	623817050926/project-4-ami
				Usage operation	RunInstances
				Virtualization type	hvm

STEP – 4: CONFIGURE GITHUB REPOSITORY □

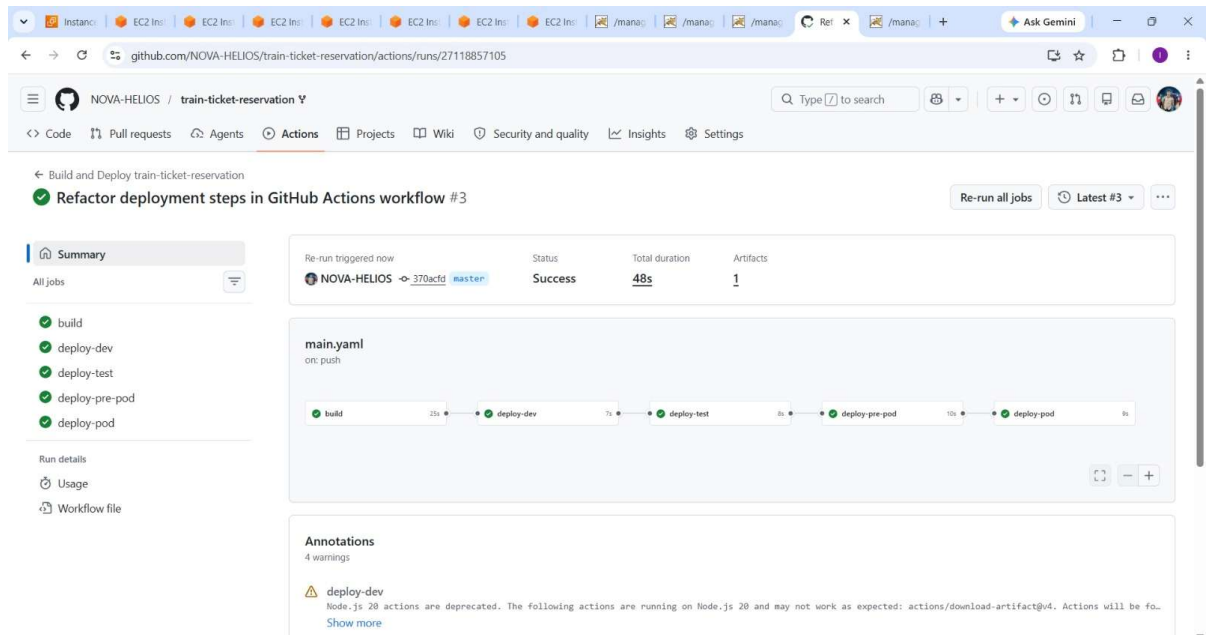
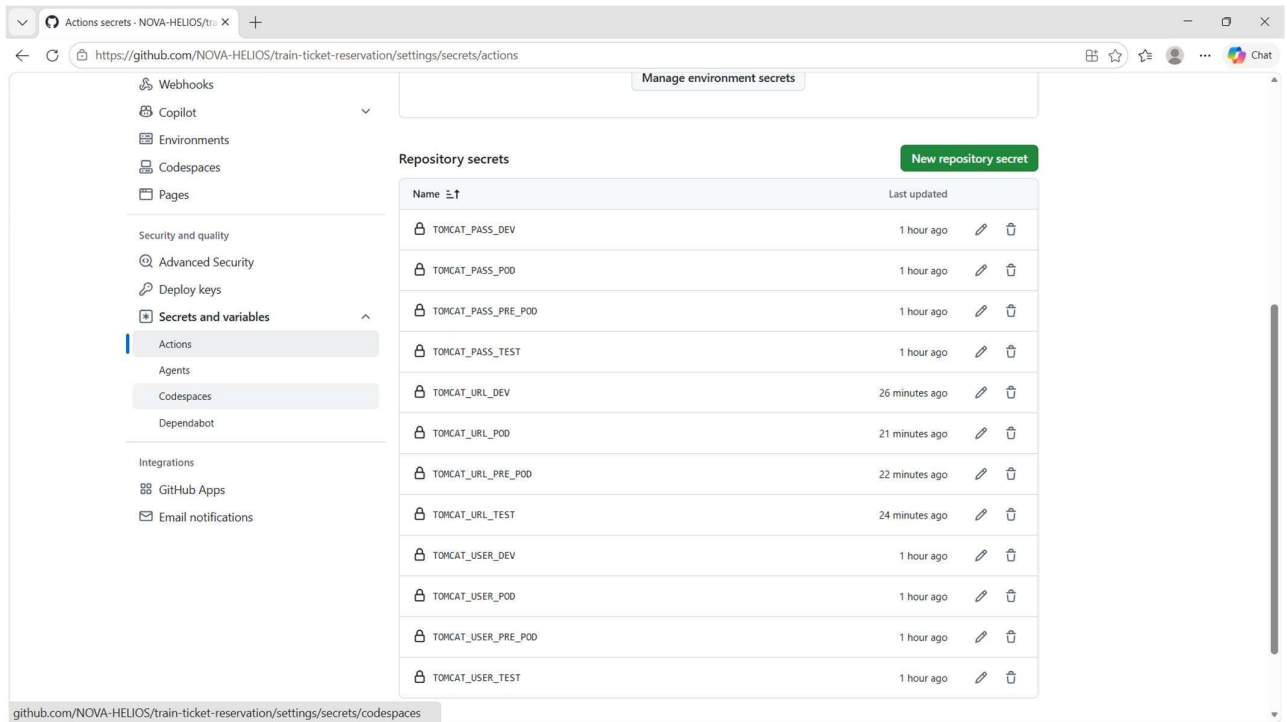
Source code was pushed to a GitHub repository.

- Create .github/workflows/shiva.yaml file was created to define the CI/CD pipeline.
- Add Secrets actions repository for CI/CD pipeline.



Step – 5: CI/CD PIPELINE USING GITHUB ACTIONS □ The pipeline checks out the code.

- ❑ Sets up Java and runs Maven commands (validate, compile, test, package).
- ❑ Uploads the WAR file as an artifact.
- ❑ Downloads the WAR in deploy jobs and uses curl to deploy to Tomcat using the Manager API.



Instance: EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins /mana /mana /mana Refacto /m: x + Ask Gemini

Not secure 16.112.70.212:8080/manager/html

Tomcat Web Application Manager

Message: OK

Manager

List Applications HTML Manager Help Manager Help Server Status

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/train-ticket-test	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Instance: EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins EC2 Ins /mana /mana /mana Refacto /mana: + Ask Gemini

Not secure 18.60.62.132:8080/manager/html

Tomcat Web Application Manager

Message: OK

Manager

List Applications HTML Manager Help Manager Help Server Status

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/train-ticket-orepod	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Instance: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: /mana: /m: x /mana: Refacto: /mana: + Ask Gemini

← → ↻ ⚠ Not secure 18.61.126.50:8080/manager/html

Tomcat Web Application Manager

Message: OK

Manager

[List Applications](#) [HTML Manager Help](#) [Manager Help](#) [Server Status](#)

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/train-ticket	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

31°C Mostly sunny 11:36 08-06-2026

Instance: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: EC2 Ins: /mana: /m: x /mana: Refacto: /mana: + Ask Gemini

← → ↻ ⚠ Not secure 16.112.109.217:8080/manager/html

Tomcat Web Application Manager

Message: OK

Manager

[List Applications](#) [HTML Manager Help](#) [Manager Help](#) [Server Status](#)

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/train-ticket-dev	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/train-ticket-reservation	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

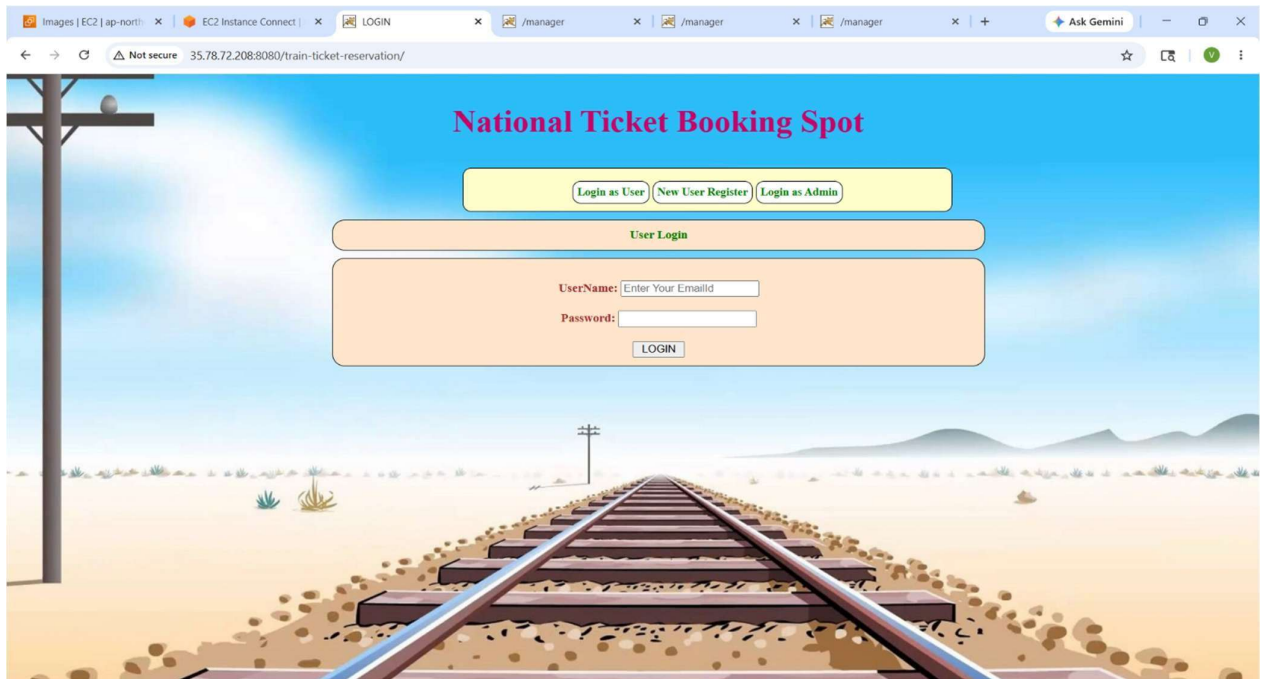
Deploy

Deploy directory or WAR file located on server

31°C Mostly sunny 11:35 08-06-2026

STEP – 8: Output

- The application was successfully built and deployed from GitHub Actions to the Tomcat servers on EC2.



- After deployment, the application was accessible via the browser using the configured context path
- The process removed manual WAR upload and made deployments faster and repeatable.